

N4 Rocket Recovery System

Continuous Assessment Presentation

Glenn Omondi

March 2026

Nakuja Project — Final Year Project

Objectives

Main Objective

To design and fabricate a reliable dual recovery system for the N4 rocket with extended telemetry range.

Specific Objectives:

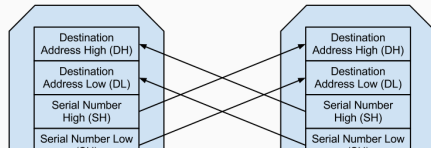
1. To extend the telemetry range of the recovery system beyond 5 km.
2. To improve deployment reliability through controlled pyrotechnic ignition.
3. To design a rugged 3D-printed avionics holder with vibration isolation, battery retention, and real-time system diagnostics.
4. To validate the complete recovery system through ground and field testing.

Design-Build-Test Iteration:

1. Define subsystem requirements from mission constraints
2. Generate mechanical, electrical, and software concepts
3. Select concepts on performance & manufacturability
4. Build CAD and circuit designs
5. Implement embedded firmware architecture
6. Execute staged bench and system-level validation

Five Implementation Modules:

- Telemetry link & antenna
- Pyrotechnic deployment & ignition control
- Power & feedback instrumentation
- Mechanical containment & avionics structure
- Embedded software & FSM



Methodology — Telemetry & Ignition Design Decisions

Telemetry — XBee-PRO 900HP

selected:

- 900 MHz ISM band, up to +24 dBm Tx
- Link budget at 5 km:
FSPL $\approx$ 105.7 dB
- Received power $\approx$ -77.7 dBm vs sensitivity -110 dBm
- Link margin: +32 dB — robust margin for flight attitude variation
- UART transparent mode — direct ESP32 integration

Ignition — PWM over DC dump:

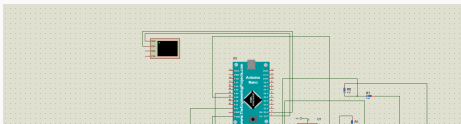
- Previous DC dump: burnout in $\sim$ 2 s (single use)
- 26 AWG nichrome: $R = 8.5 \Omega/\text{m}$
- PWM at 1 kHz with duty-cycle ramp
- Target: $D = 0.40 \rightarrow V_{\text{eff}} \approx 6 \text{ V}$ from 15 V bus
- 30 s withstand at minimum duty — confirms controlled thermal approach
- IRLZ44N MOSFET: logic-level gate, $R_{DS(\text{on})} = 22 \text{ m}\Omega$

Power & Continuity Monitoring:

- ADS1115 16-bit ADC over I²C
- 47 k Ω /10 k Ω divider scales 14–15 V bus
- 14 V $\rightarrow$ ADC code 19,650 15 V $\rightarrow$ 21,060
- Clear digital margin for state discrimination
- Launch inhibit triggered if battery <14.0 V or continuity absent

Mechanical Constraints:

- 95 mm avionics bay envelope
- Four 18650 cells in parallel (4S, 14.8 V)
- Dual threaded rods at 50 mm spacing
- **Stepped battery-carrier** — edge cells raised to clear rod paths
- PETG material for printability and chemical resistance
- Upper platform: XBee extension + beacon antenna mounting



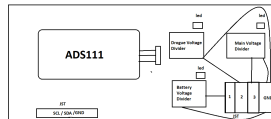
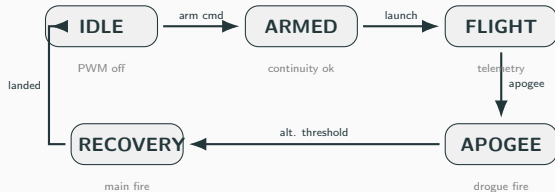
Methodology — Software Architecture (FSM)

FreeRTOS Firmware Modules:

- Sensor task (GPS, IMU, barometer) @ 10 Hz
- Telemetry TX via XBee UART @ 10 Hz
- State machine task @ 20 Hz
- Deployment channel (event-driven)
- ADS1115 monitoring @ 1 Hz

Deployment Arm Conditions:

- Battery >14.0 V
- Continuity ADC count >1000
- FSM state = APOGEE or



Methodology — Circuit Development Workflow

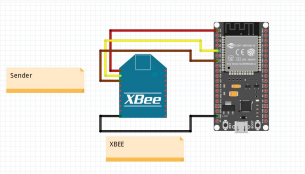
Staged development:

1. Breadboard wiring validation
2. Fritzing integration drafting
3. EasyEDA schematic + PCB capture
4. Proteus simulation
5. Bench verification

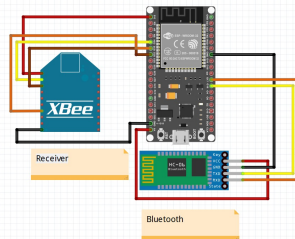
Interface strategy:

UART for XBee (high-rate stream)

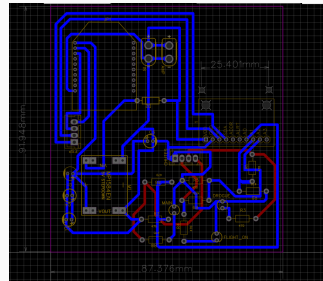
I²C for ADS1115 (monitoring)



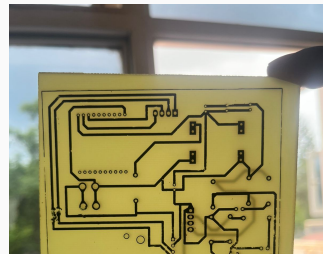
Fritzing rocket wiring



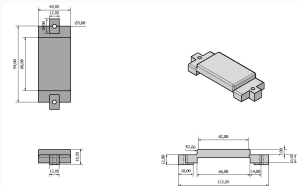
Fritzing base station



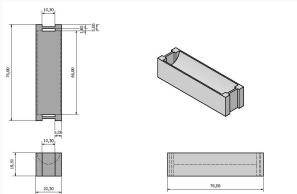
EasyEDA schematic



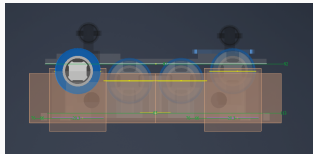
Results — Mechanical Design Outputs



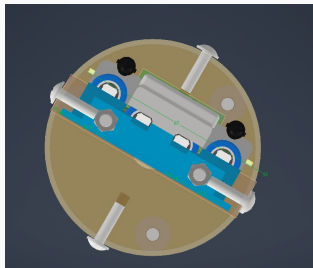
18650 case cover



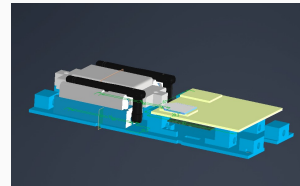
Edge battery carrier (raised)



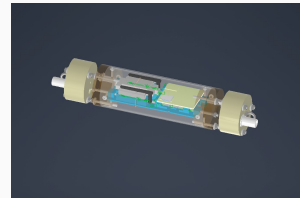
Threaded rod integration



Cross-section view



Avionics bay assembly

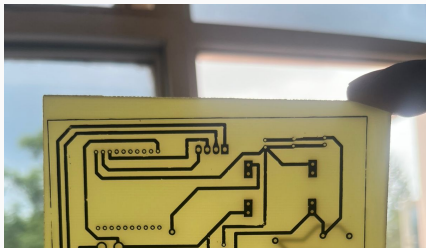


Full integration in airframe

Results — Electrical Design Outputs

PCB Fabrication (Custom):

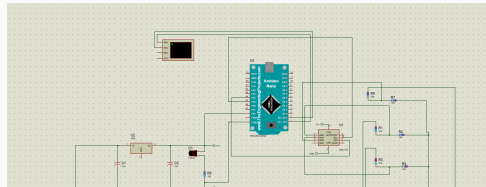
- 65 mm × 65 mm, 2-layer FR-4
- ESP32 DevKit, XBee UART interface
- 2× IRLZ44N MOSFET ignition channels
- ADS1115 monitoring circuit
- Chemical etching complete



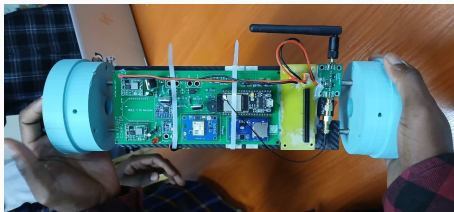
Proteus BMS Simulation:

Simulated ADS1115 divider network before PCB bring-up:

- 14 V $\rightarrow V_{ADC} = 2.456$ V (code 19,650)
- 15 V $\rightarrow V_{ADC} = 2.632$ V (code 21,060)
- Distinct codes confirm reliable state discrimination



Results — Assembled Hardware & Firmware Status



Assembled avionics bay

Firmware modules complete:

- ✓ XBee UART telemetry (10 Hz)
- ✓ GPS NMEA parsing
- ✓ IMU (MPU6050) & barometer (BMP280)
- ✓ ADS1115 voltage + continuity monitoring
- ✓ PWM ignition channel with duty ramp
- ✓ Flight state machine
(IDLE→ARMED→FLIGHT→APOGEE→)
- ✓ FreeRTOS task scheduler

XBee configured via XCTU

Pending Design & Validation Work

Hardware Pending

- PCB population and component soldering
- 3D-printed structural parts (PETG) — in production
- Aluminium explosion cap holders — pending stock
- Full avionics bay integration and fitment check

Testing Pending

- Open-field XBee range test (0.5–5 km, packet-loss measurement)

Firmware Pending

- Hardware-in-the-loop FSM validation
- SD card data logging module
- Ground-station command software interface

Task	N			D		J		F		M		A		M		J	
Proposal presentation																	
Continuous presentation																	
Literature review																	
FC Holder design																	
Xbee Circuit Design																	
Software																	
Interim report																	
Material requisition																	
PCB Fabrication																	
Range & Drone																	

Thank you

Questions?